

QCL7350 – Quantum Inspired Optimization

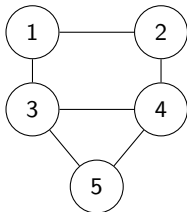
Lab 2 – Minimum Vertex Cover and Graph Coloring Problems

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Minimum Vertex Cover: Brute Force



$V = \{1, 2, 3, 4, 5\}$, $|E| = 6$

Edges:

$\{1, 2\}, \{1, 3\}, \{2, 4\}, \{3, 4\}, \{3, 5\}, \{4, 5\}$

Goal: smallest set $S \subseteq V$ such that every edge has at least one endpoint in S .

Brute force idea: try all subsets of increasing size until one covers every edge.

Try $|S| = 1$: max degree is 3 (vertex 3 or 4) $\Rightarrow$ covers at most 3 of 6 edges. **Fails.**

Try $|S| = 2$: best pair $\{3, 4\}$ covers $\{1, 3\}, \{2, 4\}, \{3, 4\}, \{3, 5\}, \{4, 5\}$ (5 edges) but misses $\{1, 2\}$. **Fails.**

Try $|S| = 3$: $S = \{2, 3, 4\}$ covers all 6 edges. **Succeeds.**

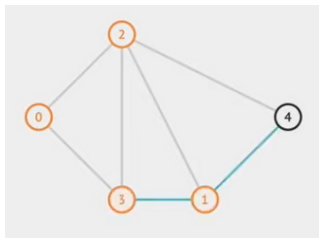
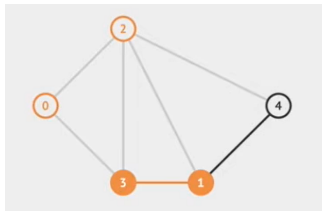
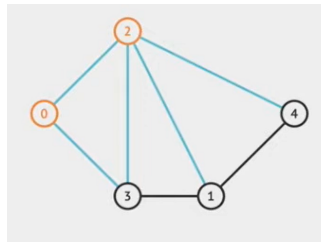
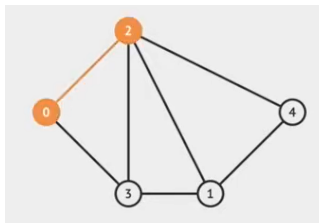
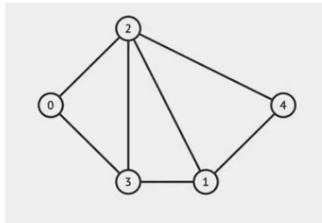
Minimum Vertex Cover = $\{2, 3, 4\}$, size = 3.

Complexity: checking all subsets is $O(2^n)$ – exponential.

Minimum Vertex Cover: 2-Approximation Algorithm

Algorithm:

1. Start with $C = \emptyset$ and $E' = E$.
2. While $E' \neq \emptyset$: pick any edge $(u, v) \in E'$, add u, v to C , remove from E' all edges touching u or v .
3. Return C . Guarantee: $|C| \leq 2 \cdot |C_{OPT}|$.

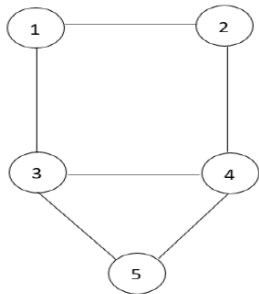


Classical Constraints and Equivalent Penalties

Classical Constraint	Equivalent Penalty
$x + y \leq 1$	$P(xy)$
$x + y \geq 1$	$P(1 - x - y + xy)$
$x + y = 1$	$P(1 - x - y + 2xy)$
$x \leq y$	$P(x - xy)$
$x_1 + x_2 + x_3 \leq 1$	$P(x_1x_2 + x_1x_3 + x_2x_3)$
$x = y$	$P(x + y - 2xy)$

This table converts constrained models into unconstrained penalty terms, the key step in building a QUBO model.

MVC: QUBO Formulation (1)



$$V = \{1, 2, 3, 4, 5\}, |E| = 6$$

Let $x_j = 1$ if vertex j is in the cover, $x_j = 0$ otherwise.

Standard constrained model:

$$\text{Minimize } \sum_{j \in V} x_j \quad \text{subject to } x_i + x_j \geq 1 \quad \forall (i, j) \in E$$

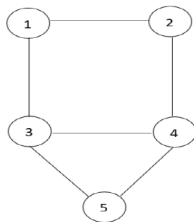
Each edge gives a constraint of the form $x + y \geq 1$, whose penalty is $P(1 - x - y + xy)$.

Unconstrained (QUBO) model:

$$\text{Minimize } y = \sum_{j \in V} x_j + P \sum_{(i, j) \in E} (1 - x_i - x_j + x_i x_j)$$

The objective rewards small covers; the penalty punishes any edge left uncovered ($x_i = x_j = 0$).

MVC: QUBO Formulation (2) – Numeric Example



$m = 5$ nodes
 $n = 6$ edges

Using the same graph as before ($m = 5$ nodes, $n = 6$ edges):

$$\begin{aligned}
 \text{Minimize } y = & x_1 + x_2 + x_3 + x_4 + x_5 \\
 & + P(1 - x_1 - x_2 + x_1x_2) + P(1 - x_1 - x_3 + x_1x_3) \\
 & + P(1 - x_2 - x_4 + x_2x_4) + P(1 - x_3 - x_4 + x_3x_4) \\
 & + P(1 - x_3 - x_5 + x_3x_5) + P(1 - x_4 - x_5 + x_4x_5)
 \end{aligned}$$

which simplifies to

$$\begin{aligned}
 \text{Minimize } y = & (1-2P)x_1 + (1-2P)x_2 + (1-3P)x_3 + (1-3P)x_4 + (1-2P)x_5 \\
 & + Px_1x_2 + Px_1x_3 + Px_2x_4 + Px_3x_4 + Px_3x_5 + Px_4x_5 + 6P
 \end{aligned}$$

Choosing $P = 8$ and dropping the constant $6P = 48$: $QUBO : \min x^t Q x$

$$Q = \begin{pmatrix} -15 & 4 & 4 & 0 & 0 \\ 4 & -15 & 0 & 4 & 0 \\ 4 & 0 & -23 & 4 & 4 \\ 0 & 4 & 4 & -23 & 4 \\ 0 & 0 & 4 & 4 & -15 \end{pmatrix}$$

Optimal solution: $x_2 = x_3 = x_4 = 1$, others 0 $\Rightarrow$ cover = $\{2, 3, 4\}$, size 3 (matches brute force).

Graph Coloring: Brute Force and Greedy Approaches

Problem: assign one of k colors to every vertex so that no two adjacent vertices share a color.

Brute Force:

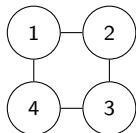
- ▶ Enumerate every possible assignment of k colors to the n vertices (k^n possibilities).
- ▶ For each assignment, check every edge: reject if both endpoints share a color.
- ▶ Keep the first (or smallest- k) feasible assignment found.
- ▶ Complexity: $O(k^n)$ – exponential, impractical for large graphs.

Greedy Approach:

- ▶ Order the vertices (e.g., $1, 2, \dots, n$).
- ▶ For each vertex, assign the smallest-numbered color not already used by its already-colored neighbors.
- ▶ Complexity: $O(n + m)$ – fast, but not always optimal; the number of colors used can depend on vertex ordering.

Example (4-cycle $1-2-3-4-1$, order $1, 2, 3, 4$): greedy assigns colors $1, 2, 1, 2$ – a valid, optimal 2-coloring.

Graph Coloring: QUBO Formulation (1) – Setup



4-cycle graph

$|V| = 4$, $|E| = 4$, use $K = 2$ colors

Let $x_{i,p} = 1$ if node i gets color p .

Assignment constraints (one color per node):

$$x_{i,1} + x_{i,2} = 1, \quad i = 1, 2, 3, 4$$

$\Rightarrow$ Each node must be assigned exactly one of the two colors.

Adjacency constraints (no two adjacent nodes share the same color):

$$x_{i,c} + x_{j,c} \leq 1, \quad c = 1, 2 \text{ for each edge } (i, j)$$

$\Rightarrow$ Adjacent nodes cannot both take the same color.

Edges: $\{1, 2\}, \{2, 3\}, \{3, 4\}, \{4, 1\}$: 4 edges $\times$ 2 colors = 8 adjacency constraints, plus 4 assignment constraints.

Relabel with a single subscript:

$$(x_{11}, x_{12}, x_{21}, x_{22}, x_{31}, x_{32}, x_{41}, x_{42}) = (x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8)$$

Graph Coloring: QUBO Formulation (2) – Assignment Penalties

Using $P(x_1 + x_2 - 1)^2$ becomes $P(-x_1 - x_2 + 2x_1x_2) + P$ (Transformation #1), applied to each node:

$$P(x_1 + x_2 - 1)^2 \Rightarrow P(-x_1 - x_2 + 2x_1x_2) + P$$

$$P(x_3 + x_4 - 1)^2 \Rightarrow P(-x_3 - x_4 + 2x_3x_4) + P$$

$$P(x_5 + x_6 - 1)^2 \Rightarrow P(-x_5 - x_6 + 2x_5x_6) + P$$

$$P(x_7 + x_8 - 1)^2 \Rightarrow P(-x_7 - x_8 + 2x_7x_8) + P$$

Taking $P = 4$, this gives the block-diagonal "developing" Q matrix (constant $4P = 16$ added separately):

$$Q = \begin{pmatrix} -4 & 4 & 0 & 0 & 0 & 0 & 0 & 0 \\ 4 & -4 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -4 & 4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 4 & -4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -4 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 4 & -4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -4 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 & 4 & -4 \end{pmatrix}$$

Each 2×2 block enforces that node i receives exactly one of the two colors.

Graph Coloring: QUBO Formulation (3) – Adjacency Penalties & Solution

For each edge and each color, insert penalty $Px_{i,p}x_{j,p}$ (Transformation #2). With $P = 4$, insert value $P/2 = 2$ symmetrically. Edges $\{1, 2\}, \{2, 3\}, \{3, 4\}, \{4, 1\}$ give the completed matrix:

$$Q = \begin{pmatrix} -4 & 4 & 2 & 0 & 0 & 0 & 2 & 0 \\ 4 & -4 & 0 & 2 & 0 & 0 & 0 & 2 \\ 2 & 0 & -4 & 4 & 2 & 0 & 0 & 0 \\ 0 & 2 & 4 & -4 & 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 & -4 & 4 & 2 & 0 \\ 0 & 0 & 0 & 2 & 4 & -4 & 0 & 2 \\ 2 & 0 & 0 & 0 & 2 & 0 & -4 & 4 \\ 0 & 2 & 0 & 0 & 0 & 2 & 4 & -4 \end{pmatrix} \quad QUBO : \min x^t Q x + 16$$

Solving this model yields the feasible coloring:

$$x_1 = x_4 = x_5 = x_8 = 1, \text{ all other variables} = 0$$

Switching back: node 1 $\rightarrow$ color 1, node 2 $\rightarrow$ color 2, node 3 $\rightarrow$ color 1, node 4 $\rightarrow$ color 2.

This is a valid, optimal 2-coloring of the 4-cycle (matches greedy result); the QUBO objective value is 0, its minimum.